Deliverable #1 : Software Requirement Specification (SRS)

SE 3A04: Software Design III – Large System Design

Tutorial Number: T02
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1 Introduction

This document presents the Software Requirements Specification (SRS) for the Smart City Environmental Monitoring and Alert System (SCEMAS). It describes the functional and non-functional requirements for a cloud-native IoT software platform that collects, processes, and acts upon environmental sensor data to support city operations and public awareness.

1.1 Purpose

The purpose is to define the software requirements for SCEMAS in sufficient detail to guide architectural design, implementation, and verification. This document serves as a contractual reference between the development team and the course instructor for evaluating project deliverables. The intended audience includes SCEMAS stakeholders, including project managers, developers, domain experts, SCEMAS team members, or investors. No prior readings are required.

1.2 Scope

The Smart City Environmental Monitoring and Alert System (SCEMAS) is a cloud-native IoT software platform that continuously collects environmental sensor telemetry, processes and validates incoming data, and produces actionable intelligence for city operators and the public. The platform ingests data streams from a distributed network of simulated sensors reporting air quality, noise levels, temperature, and humidity measurements across predefined geographical zones within a city.

The system includes four core functionalities:

1. **Telemetry Ingestion and Processing:** Sensors publish environmental readings via the MQTT protocol. The system validates each incoming message for schema conformance and plausible value ranges, persists validated data in a time-series database, and computes real-time aggregations (e.g., 5-minute averages, hourly maximums) per geographical zone.

2. **Rule-Based Alerting:** System administrators define threshold and anomaly-based alert rules. The alerting engine evaluates incoming telemetry against active rules in near real-time, triggers alerts upon detecting violations, logs for each event with full context, and notifies subscribed external systems through a dedicated API endpoint. A complete alert history with status tracking (active, acknowledged, resolved) is maintained and queryable.

3. **Data Access and Visualization:** City operators access a secure, role-authenticated dashboard displaying geographical maps of sensor locations, charts of environmental metrics, gauge indicators for current readings, system health status, and active alerts. For public and third-party consumption, the system exposes a read-only REST API serving aggregated, non-sensitive environmental data such as the current Air Quality Index for a given zone. A functional client simulating a public digital signage display demonstrates consumption of this API.

4. **Security and Administration:** All entities interacting with the system (IoT devices and human users) must complete strong authentication before any data exchange or access is permitted. A role-based access control model distinguishes between standard operators and system administrators. An audit log records all significant events, including device lifecycle changes, alert triggers, and user management actions.

The primary objective of SCEMAS is to replace sparse, manual, or siloed environmental monitoring with an integrated, real-time platform that enables faster detection of and response to pollution events, heatwaves, noise disturbances, and other environmental anomalies. The system aims to reduce operator response latency by surfacing threshold violations as structured alerts rather than requiring manual inspection of raw data streams. A secondary objective is public transparency, where aggregated environmental data is made available through the public API so that citizens and third-party developers can build on top of the platform

data.

One innovative feature includes personalized alert subscriptions for city operators. This feature allows individual operators to subscribe to specific environmental metrics, geographical zones, or alert severity levels, filtering dashboard notifications to show only the alerts relevant to their responsibilities. The goal is to reduce alert fatigue and improve per-operator response efficiency in multi-zone deployments.

Physical sensor hardware is simulated for development and testing purposes. The project focuses on architectural design and core implementation of backend services, data pipelines, and APIs. Production-grade cloud deployment, integration with real municipal data systems or third-party weather services, and mobile application development are outside the scope of this project.

1.3 Definitions, Acronyms, and Abbreviations

- **Alert Fatigue:** A condition in which operators become desensitized to frequent alerts, leading to slower response times or missed critical events.
- **Anomalous Trend:** A pattern in sensor data that deviates significantly from expected behaviour over time, potentially indicating a developing environmental event.
- **AQI:** Air Quality Index, a standardized numerical scale for reporting air quality, is calculated from pollutant concentrations.
- **At-Least-Once Delivery:** A messaging guarantee in which every message is delivered to the consumer at least one time, though duplicates may occur.
- **Audit Log:** A chronological record of significant system events (e.g., authentication attempts, alert triggers, configuration changes) used for security monitoring and compliance.
- **Cloud-Native:** Software designed to run on cloud infrastructure, leveraging containerization, microservices, and scalable resource management.
- **Digital Signage:** A public-facing electronic display used to present information; in this project, simulated as a functional client application.
- **Geographical Zone:** A predefined administrative region within the city used for aggregating and reporting environmental data. Zones do not resolve precise locations, in accordance with privacy-by-design principles.
- **Horizontal Scaling:** A scaling strategy in which additional instances of a service are deployed to distribute load, as opposed to increasing the resources of a single instance (vertical scaling).
- **IoT:** The Internet of Things is a network of interconnected sensors and devices that collect and exchange data.
- **JSON:** JavaScript Object Notation, a lightweight data interchange format used for sensor telemetry messages.
- **MQTT:** Message Queuing Telemetry Transport is a lightweight publish-subscribe messaging protocol commonly used in IoT systems.
- **P95:** The 95th percentile of a latency distribution; 95% of requests complete within the stated time.
- **PII:** Personally Identifiable Information is any data that could be used to identify a specific individual. SCEMAS shall not collect, process, or store PII.
- **$PM_{2.5}$:** Particulate Matter with a diameter less than 2.5 micrometres, a key air quality indicator

measured in $\mu g/m^3$.

- **Privacy-by-Design:** A systems engineering approach in which privacy protections are embedded into the architecture from the outset, rather than added retroactively.
- **Rate limiting:** A mechanism that restricts the number of API requests a client can make within a given time window, used to prevent abuse and ensure fair access.
- **RBAC:** Role-Based Access Control is an authorization model that assigns permissions based on user roles (e.g., operator, administrator).
- **REST API:** Representational State Transfer Application Programming Interface is a standardized architectural style for web service communication.
- **Telemetry:** Automated measurement data are transmitted from remote sensors to the system for processing and storage.
- **Threshold Violation:** An event in which a sensor reading exceeds or falls below a predefined acceptable range, triggering an alert.
- **Time-Series Database:** A database optimized for storing and querying timestamped data points, such as InfluxDB or TimescaleDB.
- **TLS:** Transport Layer Security is a cryptographic protocol for encrypting data in transit. Version 1.3 (RFC 8446) is the current standard.
- **WCAG:** Web Content Accessibility Guidelines is a set of recommendations for making web content more accessible, published by the W3C.

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1.5 Overview

Section 2 provides an overall product description, including the product's context, major functions, user characteristics, constraints, assumptions, and deferred requirements. Section 3 presents the use-case diagram for the system. Section 4 specifies the functional requirements as use cases organized by business events and viewpoints, culminating in global scenarios. Section 5 enumerates the non-functional requirements across look and feel, usability, performance, operational, maintainability, security, cultural, and legal categories. Section 6 describes the innovative feature selected for implementation.

2 Overall Product Description

2.1 Product Perspective

The Smart City Environmental Monitoring and Alert System (SCEMAS) is a standalone, cloud-native IoT software platform designed to operate as a part of a broader smart city ecosystem to collect and analyze environmental sensor data to guide city operations and alert the public. It is designed to interface with external entities, including IoT sensor networks, public information displays, and third-party software consumers, through well-defined and standardized interfaces. Currently, Battair and Airly are two existing products related to SCEMAS. Bettair is a highly precise multi-pollutant sensor system that measures and maps several parameters, specifically pollution (NO_2, CO_2, etc.), to provide real-time information on pollutant levels, aiming to mitigate air pollution and improve the quality of life [27]. Similarly, Airly provides a network of sensors that map air quality and send alerts when pollution thresholds are reached, to help create clean air zones and guide traffic management. SCEMAS is similar in process; however, instead of a physical sensing system, it operates as a software-only platform that consumes environmental data produced by external sensor networks [28].

SCEMAS is largely self-contained in terms of data processing, alert evaluation, and user access control. It does not depend on external municipal databases or emergency response systems to function correctly. However, it exists within a larger smart city ecosystem and interacts with external systems through well-defined interfaces. The system interfaces with external environmental sensors using the MQTT protocol, allowing sensor devices to publish telemetry data to the platform. City operators access SCEMAS through a secure web-based dashboard that provides real-time visualizations, system health indicators, and alert management functionality. Public users and third-party developers are interested in the system through a read-only REST API. All external interactions are secured through authentication, authorization, and encrypted communication channels.

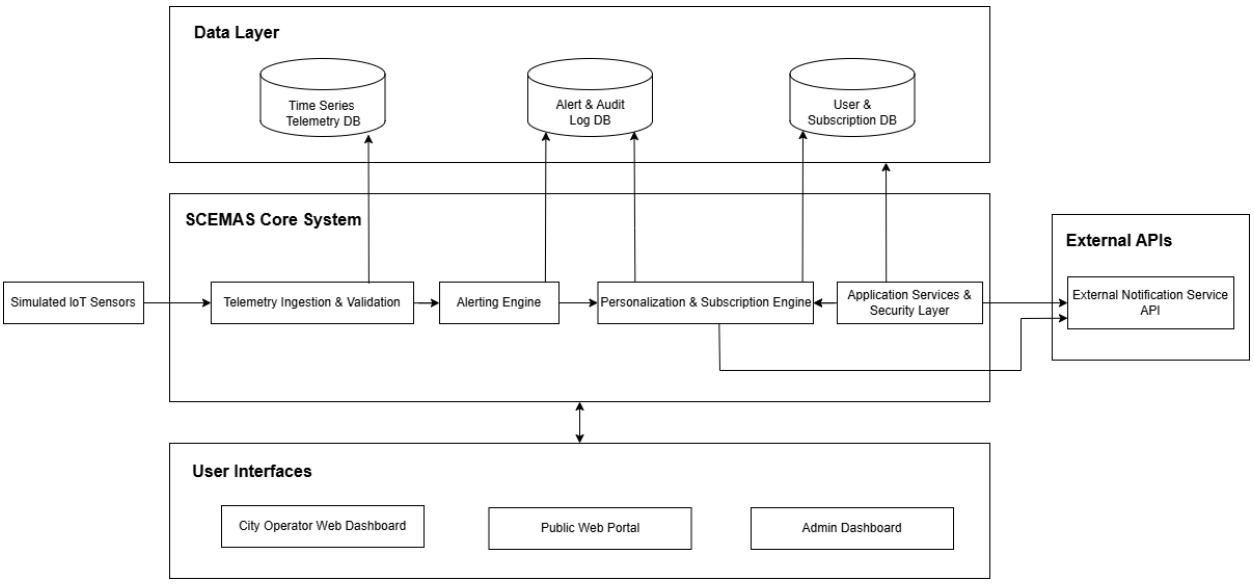


Figure 1: Block Diagram

2.2 Product Functions

There will be 4 modules in the product: Telemetry Ingestion and Processing Module, Rule-Based Alerting Engine, Data Access Module, and Security & Administration Module. The major functions of the software in the Telemetry Processing Module include ingesting sensor telemetry, validating incoming messages, and storing time-series environmental data. Within the Alerting Engine, the system supports defining alert rules, detecting rule violations in near real-time, and generating and recording alerts. The Data Access Module provides a secure operator dashboard for real-time and historical environmental monitoring and exposes a read-only REST API for aggregated, non-sensitive public data, supporting third-party data consumption. The Security and Administration Module include strong authentication for all entities, role-based access control, and maintains an audit log of significant system events.

Modules	**Functions**

Telemetry Ingestion and Processing Module	• Ingest sensor telemetry
	  - Accepts data streams via the MQTT protocol and processes incoming sensor data
	• Validate telemetry
	  - Ensures messages follow the correct format, schema, and plausible value ranges
	• Store timeseries data
	  - Saves validated telemetry in a database optimized for timeseries data

Rule-Based Alerting Engine	• Configure alert rules
	  - Allows administrators to define alert rules based on specific thresholds or anomalous trends
	• Detect rule violations
	  - Evaluates incoming sensor data in near real-time to identify alert conditions
	• Generate and record alert
	  - Triggers alerts, logs events, and updates alert status

Data Access Module	• Provide operator dashboard
	  - Displays real-time visualizations, system health, active alerts, and historical trends
	• Offer public data access
	  - Exposes a read-only REST API for aggregated, nonsensitive environmental data
	• Support third-party consumption
	  - Enables external clients, such as digital signage, to retrieve environmental data

Security and Administration Module	• Authenticate all entities
	  - Requires strong authentication for IoT devices and human users before access is permitted
	• Manage access roles
	  - Uses a role-based access control (RBAC) model to distinguish operators and administrators
	• Maintain audit logs
	  - Records significant events, including device lifecycle changes, alert triggers, and user management actions

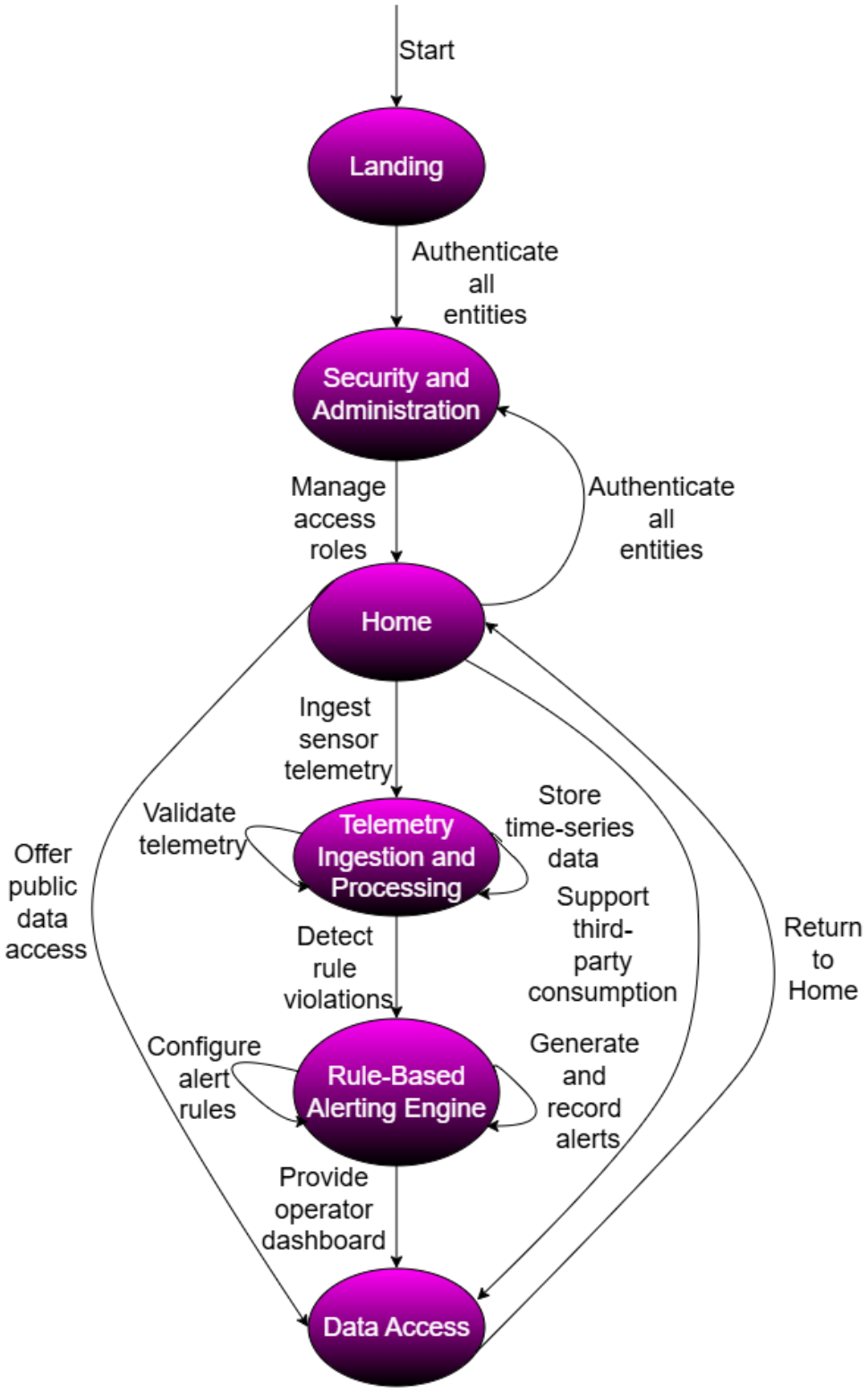


Figure 2: State Diagram

2.3 User Characteristics

1. Education

   - City Operators: Operators only need basic reading and writing skills, along with a general sense of how environmental conditions affect city operations.
   - Public Users: Public users just need basic literacy. They should be able to read short text, look at simple charts, and understand basic numbers.
   - Third-Party developers: Third-party developers usually have an education in computer science, software engineering, or a related technical field. They are expected to understand software development concepts such as APIs, data formats, and how web services communicate.

2. Experience

   - City Operators: Some operators may have used dashboards or monitoring tools before, but the system is designed so that even someone new can figure out the main features, like checking sensor data or acknowledging alerts without much help.
   - Public Users: Public users may have no prior experience with environmental monitoring tools. The public-facing interface is intentionally simple, requiring only basic interactions such as reading displayed information or requesting data through the client.
   - Third-Party Developers: Many third-party developers will already have experience working with public APIs or integrating external data into applications. They may use SCEMAS data to build tools such as environmental dashboards, research applications, or other software that displays or analyzes environmental conditions.

3. Technical Expertise

   - City Operators: Operators should be comfortable with basic computer use, like navigating a web page, clicking through menus, and reading charts or maps.
   - Public Users: Public users only need basic comfort with looking at information on a screen, whether that's on a phone, computer, or digital display.
   - Third Party-Developers: Third-party developers are expected to have a strong technical background and be comfortable working with APIs, JSON data, and common programming tools. They should be familiar with concepts such as HTTP requests, API key authentication, and general software development workflows.

2.4 Constraints

1. **Budget:** The project operates under a $0 budget, which means that our development must rely exclusively on free educational tiers of cloud services, open-source software, and publicly available tools.

2. **Time and Expertise:** SCEMAS is developed as a part of an academic course project, which imposes strict constraints on developmental time, team size, and scope.

3. **Hardware and Sensor Limitations:** The project does not use real hardware. All sensor data must come from scripts or simulated devices.

2.5 Assumptions and Dependencies

1. Environmental sensor data is generated by simulated IoT devices which accurately reflect the structure and frequency of real-world sensor telemetry.
2. Each environmental sensor is assumed to monitor one primary geographic zone, and sensor-to-zone mappings do not change dynamically during system operation.
3. The operator dashboard will be accessed through a modern web browser that supports the required charts and visual elements.

4. The system will only be used within Canada, so all timestamps, units, and conventions assume a Canadian context (e.g., Eastern Time, metric units).
5. Public access to environmental data is limited to aggregated, non-sensitive information, and public users are not permitted to access raw telemetry or system configuration data.
6. City operators interact with the system exclusively through a web-based dashboard, and no mobile-native operator application is required.
7. Third-party systems and developers access SCEMAS functionality only through the exposed read-only REST API, and no direct database or internal service access is permitted.

Other assumptions that, if it fails to hold, could require a change to the requirements:

1. The system is expected to run on the latest versions of common web browsers. If older browsers must be supported, UI and performance requirements may need to change.

2.6 Apportioning of Requirements

1. Integration with physical environmental sensor hardware
   - Future versions of SCEMAS may support direct integration with real-world IoT sensor devices deployed across a city.
2. Support for multiple cities or regions
   - Future versions may extend support to multiple cities, regions, or jurisdictions, requiring changes to data models and system configuration.
3. Mobile applications for operators or public users
   - Future releases may include native mobile applications for city operators or public users to access environmental data and alerts. The current system provides access through web-based dashboards and APIs only.
4. Multi-language support
   - The initial release is in English, and additional languages can be added in later releases.
5. Automated long-term analytics
   - Features like forecasting, anomaly detection, or trend prediction are not included in the first release but will be added later.

3 Use Case Diagram

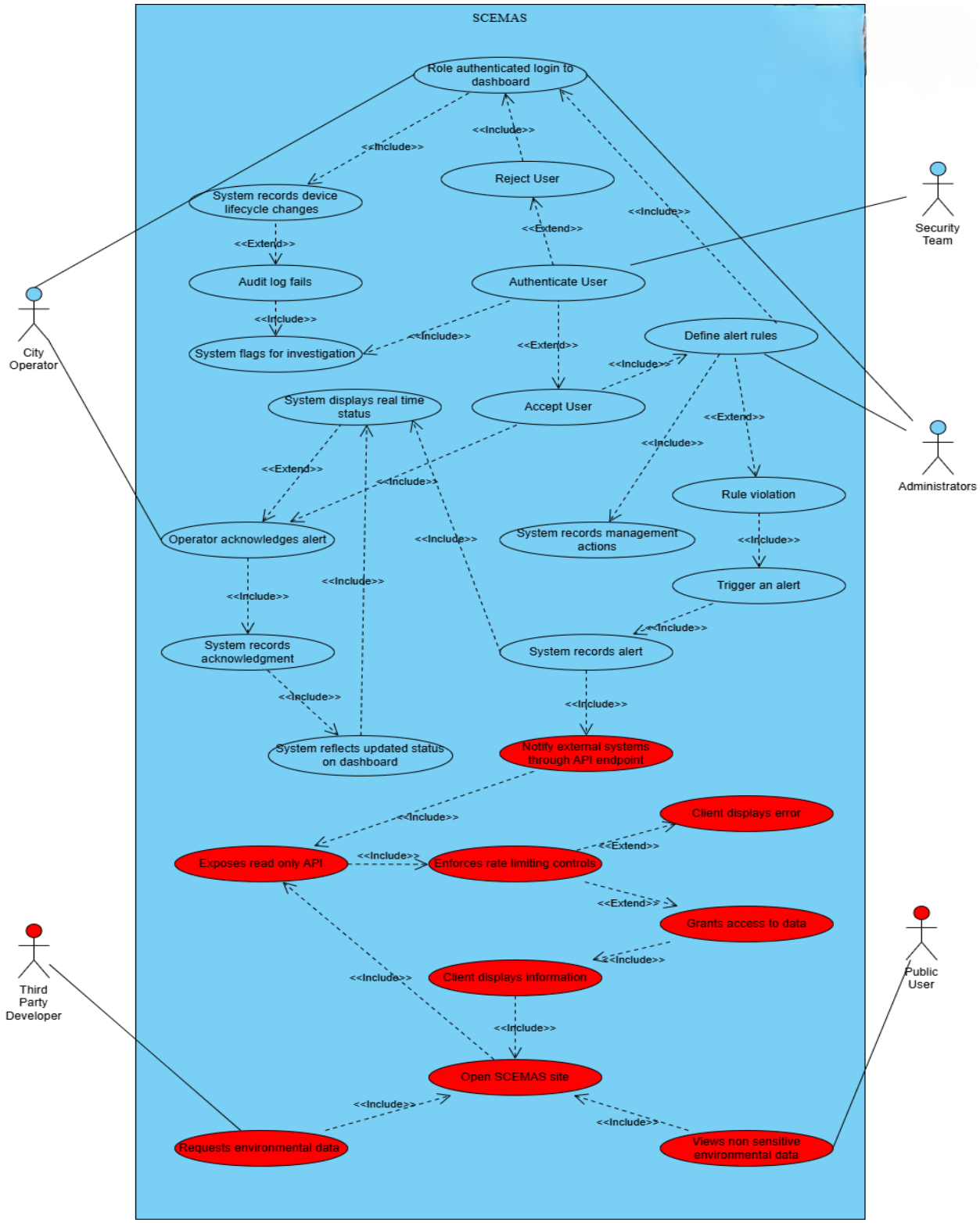


Figure 3: Use Case Diagram

4 Highlights of Functional Requirements

The business events we will consider include:

- BE1 Signup and Login
- BE2 Request and Interpret Environmental Data
- BE3 Define Alert Rules
- BE4 Acknowledge a Critical Environmental Alert
- BE5 Manage Personalized Alert Subscriptions

The Viewpoints we will consider are:

- VP1 City Operators
- VP2 System Administrators
- VP3 Public Users
- VP4 Security and Accountability Team
- VP5 Third-Party Developers

Business Events:

BE1. Signup and Login #1

Pre-Condition: The user or external system must have valid credentials. The SCEMAS system is operational and ready to accept authentication requests.

VP1. City Operators #1

Main Success Scenario

1. User opens the SCEMAS dashboard.
2. System displays signup and login options.
3. User selects either the signup or the login option.
4. For signup: user provides required signup details; system creates an account and assigns the City Operator role.
5. For Login: user enters credentials; system verifies credentials against the RBAC model.
6. If correct, the system grants access and displays dashboard operations for City Operators.

Secondary Scenario

4i. Invalid signup information.

4i.1. Submitted signup details are invalid.

4i.2. System displays error message; prompts retry or contact the Security and Accountability Team.

4ii. Incorrect login credentials.

4ii.1. Credentials are invalid.

4ii.2. System displays error message; prompts retry or contact the Security and Accountability Team.

4iii. System technical failure.

4iii.1. Signup or login cannot be processed.

4iii.2. System displays an error message and logs failure.

VP2. System Administrators #2
NA

VP3. Public Users #3
NA

VP4. Security and Compliance Team #4
4i. System logs all signup and login attempts (success and failure) for monitoring.
4ii. Repeated invalid login attempts occur; the Security and Accountability Team is notified.
4iii. Suspicious signup patterns trigger automated alerts for review.

VP5. Third-Party Developers #5
NA

Global Scenario
Pre-Condition: The human user (City Operator, System Administrator, Public User, Security and Accountability Team member) or an authenticated external system (e.g., third-party application or IoT gateway using an API key) has valid login credentials or provides valid signup information for account creation.
Main Success Scenario
1. Human user or external system opens the SCEMAS dashboard or API interface.
2. System displays the appropriate signup and login fields.
3. Human user enters signup details or login credentials; external system provides an API key.
4. System validates the submitted information and verifies credentials according to the RBAC model.
5. If validation succeeds, the system grants access based on the authenticated role:
   i. City Operator is granted access to dashboard operations
   ii. The System Administrator is granted access to management operations
   iii. Public User is granted access to aggregated environmental data
   iv. Security and Accountability Team is granted access to audit logs and monitoring
   v. Third-Party Developer is granted access to API endpoints

Secondary Scenario
4i. Invalid signup information or incorrect credentials/API key.
   4i.1. Submitted signup information or login credentials are invalid.
   4i.2. System displays an error message and prompts retry or contact with the Security and Accountability Team.
4ii. System technical failure.
   4ii.1. Signup or login cannot be processed.
   4ii.2. System displays an error message and logs the failure.

BE2. Request and Interpret Environmental Data #2
Pre-Condition: The SCEMAS system and API are operational. Additionally, there is an interest in environmental data from third-party users or general members of the public.

VP1. City Operators #1
NA

VP2. System Administrators #2
NA

VP3. Public Users #3
Main Success Scenario
1. User opens the SCEMAS functional client, which simulates a public digital signage display.
2. The system exposes a read-only REST API.

3. Through the API, the user requests a resource that the server has.

4. Public API makes sure there is no attempt at unauthorized data scraping or denial of service attacks.

5. If there are no security violations, the server then responds with the requested resource back to the client through the API.

6. The client then displays the requested information, in this case, aggregated, non-sensitive environmental data.

Secondary Scenario

3i. Request not processed.

3i.1. An error occurs which invalidates the request to the server.

3i.2. Display error message; prompt user to refresh page and to try again.

4i. Security violation.

4i.1. Detected potential signs of dangerous behaviour.

4i.2. Display error message; prompt retry or contact the Security and Accountability Team.

5i. Server technical failure.

5i.1. Server fails to respond with the requested resource, and nothing is sent back.

5i.2. Display error message and log failure.

VP4. Security and Compliance Team #4

4i. System recognizes potential signs of a security breach from the client side.

4ii. Repeated recognized offences occur; the Security and Accountability Team is notified.

VP5. Third-Party Developers #5

Main Success Scenario

1. Third-party developer opens the SCEMAS functional client, which simulates a public digital signage display.

2. The system exposes a read-only REST API.

3. From the client, the third-party developer follows the provided comprehensive guidelines and requests specific information through the API to be interpreted within a two-hour timeframe.

4. Public API makes sure there is no attempt at unauthorized data scraping or denial of service attacks.

5. If there are no security violations, the server then responds with the requested resource back to the client through the API.

6. The client then displays the requested information, in this case, aggregated, non-sensitive environmental data.

Secondary Scenario

3i. Request not processed.

3i.1. An error occurs which invalidates the request to the server.

3i.2. Display error message; prompt user to refresh page and to try again.

4i. Security violation.

4i.1. Detected potential signs of dangerous behaviour.

4i.2. Display error message; prompt retry or contact the Security and Accountability Team.

5i. Server technical failure.

5i.1. Server fails to respond with the requested resource, and nothing is sent back.

5i.2. Display error message and log failure.

Global Scenario

Pre-Condition: The SCEMAS system and API are operational. Additionally, there is an interest in environmental data from third-party users or general members of the public.

Main Success Scenario

1. Third-party developers or public users open the SCEMAS functional client, which simulates a public digital signage display.
2. The system exposes a read-only REST API.
3. From the client, public users may make a request to the server for a resource; alternatively, third-party developers may follow the provided comprehensive guidelines to request specific information.
4. Public API makes sure there is no attempt at unauthorized data scraping or denial of service attacks.
5. If there are no security violations, the server then responds with the requested resource back to the client through the API.
6. The client then displays the requested information, in this case, aggregated, non-sensitive environmental data.

Secondary Scenario

3i. Request not processed.
- 3i.1. An error occurs which invalidates the request to the server.
- 3i.2. Display error message; prompt user to refresh page and to try again.

4i. Security violation.
- 4i.1. Detected potential signs of dangerous behaviour.
- 4i.2. Display error message; prompt retry or contact the Security and Accountability Team.

5i. Server technical failure.
- 5i.1. Server fails to respond with the requested resource, and nothing is sent back.
- 5i.2. Display error message and log failure.

BE3. Define Alert Rules #3

Pre-Condition: The system can receive sensor data, and the user must be authenticated with System Administrator privileges.

VP1. City Operators #1

NA

VP2. System Administrators #2

Main Success Scenario

1. System Administrator accesses the alert rule management interface.
2. System displays existing alert rules and configuration options.
3. System Administrator selects to define alert rules (create or modify).
4. System Administrator specifies rule parameters based on:
   a. Environmental Metric Threshold
   b. Anomaly Condition
   c. Applicable Geographical Zone
5. System validates rule configuration.
6. System stores alert rule.
7. System activates alert rule.
8. System confirms alert rule has been successfully defined and stored.

Secondary Scenario

5i. Rule configuration fails validation.
- 5i.1. System identifies invalid or missing parameters.
- 5i.2. System prompts Administrator to correct inputs.
- 5i.3. Define Alert Rules failed.

6i. System fails to store the alert rule.
- 6i.1. Storage error occurs.
- 6i.2. System notifies Administrator.
- 6i.3. Define Alert Rules failed.

7i. System fails to activate alert rule.
7i.1. Activation error occurs.
7i.2. System notifies Administrator.
7i.3. Define Alert Rules failed.

VP3. Public Users #3
NA

VP4. Security and Compliance Team #4
Main Success Scenario
1. System records the alert rule creation/modification in audit logs.
2. Audit logs capture administrator's identity, timestamp, and rule details.
3. Security and Compliance accesses audit logs for review when required.

Secondary Scenario
1i. System fails to record the audit log entry.
1i.1. Audit logging error occurs.
1i.2. System flags the failure for investigation.

VP5. Third-Party Developers #5
NA

Global Scenario
Pre-Condition: The system can receive sensor data, and the user must be authenticated with System Administrator privileges.
Main Success Scenario
1. System Administrator accesses the alert rule management interface.
2. System displays existing alert rules and configuration options.
3. System Administrator selects to define alert rules (create or modify).
4. System Administrator specifies rule parameters based on:
   a. Environmental Metric Threshold
   b. Anomaly Condition
   c. Applicable Geographical Zone
5. System validates rule configuration.
6. System stores alert rule.
7. System activates alert rule.
8. System confirms alert rule has been successfully defined and stored.

Secondary Scenario
5i. Rule configuration fails validation.
5i.1. System identifies invalid or missing parameters.
5i.2. System prompts Administrator to correct inputs.
5i.3. Define Alert Rules failed.
6i. System fails to store the alert rule.
6i.1. Storage error occurs.
6i.2. System notifies Administrator.
6i.3. Define Alert Rules failed.
7i. System fails to activate alert rule.
7i.1. Activation error occurs.
7i.2. System notifies Administrator.
7i.3. Define Alert Rules failed.

BE4. Acknowledge a Critical Environmental Alert #4
Pre-Condition: A critical environmental alert exists with the status Active, the SCEMAS dashboard is operational, and the City Operator is authenticated and authorized to acknowledge alerts.

VP1. City Operators #1

Main Success Scenario

1. City Operator accesses the dashboard.
2. System displays real-time status, including Active Critical Alerts.
3. City Operator selects a critical alert from the list.
4. System displays alert details.
5. City Operator selects Acknowledge Alert.
6. System records the acknowledgement.
7. System confirms the alert has been successfully acknowledged.
8. System reflects the updated status on the dashboard view.

Secondary Scenario

3i. No critical alerts are available to acknowledge.
- 3i.1. Alert list contains no active critical alerts.
- 3i.2. Acknowledge Critical Environmental Alert failed.

6i. System fails to update alert status.
- 6i.1. Database or service error occurs.
- 6i.2. System notifies City Operator and retains alert status as Active.
- 6i.3. Acknowledge Critical Environmental Alert failed.

8i. System fails to refresh the dashboard state after acknowledgement.
- 8i.1. User interface update fails or connection drops.
- 8i.2. System prompts City Operator to refresh and retry.
- 8i.3. Acknowledge Critical Environmental Alert partially completed.
- 8i.4. Acknowledge Critical Alert failed.

VP2. System Administrators #2
NA

VP3. Public Users #3
NA

VP4. Security and Compliance Team #4

Main Success Scenario

1. System records the acknowledgement in the audit log, including timestamp, city operator identity, and critical event details.

Secondary Scenario

1i. Audit log write fails.
- 1i.1. System flags for investigation.

VP5. Third-Party Developers #5
NA

Global Scenario

Pre-Condition: A critical environmental alert exists with the status Active, the SCEMAS dashboard is operational, and the City Operator is authenticated and authorized to acknowledge alerts.

Main Success Scenario

1. City Operator accesses the dashboard.
2. System displays real-time status, including Active Critical Alerts.
3. City Operator selects a critical alert from the list.
4. System displays alert details.
5. City Operator selects Acknowledge Alert.
6. System records the acknowledgement.
7. System confirms the alert has been successfully acknowledged.
8. System reflects the updated status on the dashboard view.

Secondary Scenario

3i. No critical alerts are available to acknowledge.
- 3i.1. Alert list contains no active critical alerts.

3i.2. Acknowledge Critical Environmental Alert failed.

6i. System fails to update alert status.

6i.1. Database or service error occurs.

6i.2. System notifies City Operator and retains alert status as Active.

6i.3. Acknowledge Critical Environmental Alert failed.

8i. System fails to refresh the dashboard state after acknowledgement.

8i.1. User interface update fails or connection drops.

8i.2. System prompts City Operator to refresh and retry.

8i.3. Acknowledge Critical Environmental Alert partially completed.

8i.4. Acknowledge Critical Alert failed.

BE5. Manage Personalized Alert Subscriptions #5

Pre-Condition: The city operator is authenticated and has access to the operator dashboard.

VP1. City Operators #1

Main Success Scenario

1. Operator opens the SCEMAS Dashboard.
2. System verifies the operator's authentication status.
3. System displays the operator dashboard with available functions.
4. Operator accesses the alert subscription management feature.
5. System displays available environmental metrics, zones, and alert types.
6. Operator selects one or more alert categories to subscribe to.
7. System saves the operator's subscription preferences.
8. System updates alert filtering based on the operator's selections.
9. System displays a confirmation message indicating that personalized alert subscriptions have been successfully updated.

Secondary Scenario

2i. System fails to verify authentication.

2i.1. System fails to authenticate the operator.

2i.2. System denies access and displays an error message.

6i. Operator enters invalid or unsupported subscription selections.

6i.1. Operator selects an invalid metric, zone, or alert type.

6i.2. System rejects the selection and displays an error message.

7i. System fails to save subscription preferences.

7i.1. System fails to store the updated preferences due to a technical issue.

7i.2. System displays an error message and logs the failure.

VP2. System Administrators #2

NA

VP3. Public Users #3

NA

VP4. Security and Compliance Team #4

NA

VP5. Third-Party Developers #5

NA

Global Scenario

Pre-Condition: The system supports operator-specific alert subscription preferences.

Main Success Scenario

1. Environmental telemetry is processed by the alerting engine.
2. System evaluates alert rules and identifies a threshold violation.
3. System checks operator subscription preferences.

4. Only operators subscribed to the relevant alert type or zone receive the notification.
5. Operators respond more efficiently due to reduced alert noise.

Secondary Scenario

3i. Operator updates their subscription preferences.
   3i.1. System immediately applies new filtering rules.
   3i.2. Updated preferences affect future alert.

5 Non-Functional Requirements

5.1 Look and Feel Requirements

5.1.1 Appearance Requirements

LF-A1. The operator dashboard shall present environmental data through geographical maps displaying sensor locations, line/bar charts for environmental metrics, and gauge indicators for current values. All visualizations shall be legible at a viewing distance of 0.5 meters on a standard 1920x1080 display.
Rationale: A legibility standard is included because operators in a control room setting must read dashboard information without physical strain [3].

LF-A2. The public-facing functional client (simulating a digital signage display) shall present aggregated environmental data (e.g., AQI by zone) using high-contrast text and visual indicators sized for readability at 3 meters.
Rationale: Digital signage is typically viewed at a standing distance, so 3-meter readability is a standard baseline for public displays.

LF-A3. The system shall use a consistent colour scheme across all interfaces, with distinct colour-coding for alert severity levels (e.g., green for normal, yellow for warning, red for critical).
Rationale: Colour-coded severity is standard practice in monitoring dashboards and aligns with ISO 22727 guidelines for hazard communication. Consistent colour conventions reduce cognitive load during time-sensitive alert response.

LF-A4. The operator dashboard shall achieve a minimum colour contrast ratio of 4.5:1 for normal text and 3:1 for large text, in compliance with WCAG 2.1 Level AA [7].
Rationale: Municipal workplaces must accommodate employees with visual impairments. WCAG 2.1 Level AA is the accepted standard for public-sector web applications in Canada.

LF-A5. The system shall not include animations or UI elements that flash more than 3 times per second.
Rationale: Prevents triggering photosensitive epilepsy and complies with WCAG 2.1 guideline 2.3.1 [7].

5.1.2 Style Requirements

LF-S1. The operator dashboard shall follow a professional, utilitarian visual style appropriate for a municipal control room environment. The purpose of the decorative elements that do not serve an informational purpose shall be avoided.
Rationale: We imagined stakeholders (city officials and operators) would expect an interface that conveys reliability. A utilitarian style reduces visual clutter and supports sustained attention during long monitoring shifts.

LF-S2. All typography across the system shall use a sans-serif font family, with a minimum body text size of 14px for the operator dashboard and 24px for the public display client.
Rationale: Sans-serif fonts are established as more readable on screens. Minimum font sizes are set based on the intended viewing distance for each interface (0.5 m for the operator dashboard, 3 m for public display).

LF-S3. The operator dashboard shall maintain a consistent and distortion-free layout across all supported viewport sizes (minimum 1280*720). UI elements shall scale proportionately, and text shall remain legible without horizontal scrolling.
Rationale: Operators may use different monitor configurations. The system should prevent visual distortion that could cause misinterpretation of environmental data during critical events.

5.2 Usability and Humanity Requirements

5.2.1 Ease of Use Requirements

UH-EOU1. A trained city operator shall be able to log into the dashboard and successfully acknowledge a critical environmental alert within 30 seconds of the alert appearing on the dashboard.
Rationale: This is an explicit requirement from the project specification [3]. It establishes a measurable upper bound on the combined authentication and alert response workflow.

UH-EOU2. The operator dashboard shall require no more than 3 clicks (or equivalent interactions) to navigate from the main dashboard view to the detail view of any active alert.
Rationale: The 30-second acknowledgment target (UH-EOU1) implies that critical workflows must have minimal navigation depth, particularly during multi-alert scenarios.

UH-EOU3. Alert subscription preferences (the innovative feature) shall be configurable from a single interface without requiring the operator to navigate away from the dashboard.
Rationale: Personalized alert subscriptions reduce alert fatigue. If the configuration interface itself introduces navigation friction, the usability benefit is undermined.

UH-EOU4. The system shall display a progress indicator when any asynchronous operation is in progress (e.g., loading dashboard data, processing an API request).
Rationale: Progress indicators prevent user frustration by providing feedback on system status, ensuring operators know the system has not crashed or stalled [8].

5.2.2 Personalization and Internationalization Requirements

UH-PI1. The system shall support personalized alert subscriptions, allowing each city operator to subscribe to specific environmental metrics, geographical zones, or alert severity levels. The system shall filter dashboard notifications to display only alerts matching the operator's active subscriptions.
Rationale: This is the selected innovative feature. Operators responsible for different districts or concerns receive all alerts by default, leading to alert fatigue. Personalized subscriptions ensure that each operator sees only relevant alerts, improving response efficiency.

UH-PI2. The operator dashboard shall display timestamps in Eastern Time (ET) and measurements in metric units (°C, µg/m^3, dB).
Rationale: We imagined stakeholders in a Canadian municipality would expect local time conventions and metric units. Unfamiliar units introduce a translation step that slows interpretation during time-sensitive scenarios.

5.2.3 Learning Requirements

UH-L1. A competent third-party developer with no prior exposure to SCEMAS shall be able to successfully request and interpret environmental data from the public REST API within a two-hour timeframe, using only the provided API documentation.
Rationale: This is an explicit requirement from the project specification [3].

UH-L2. A new city operator shall be able to perform all core dashboard operations (view real-time data, acknowledge alerts, configure alert subscriptions) after a training session of no more than 1 hour.
Rationale: Municipal staff turnover and shift rotations require new personnel to become productive quickly. One hour is a reasonable upper bound given that the interface supports a limited set

of well-defined tasks.

5.2.4 Understandability and Politeness Requirements

UH-UP1. All error messages displayed to users shall include: (a) a description of what went wrong, (b) the likely cause, and (c) a suggested corrective action. The system shall avoid blaming language (e.g., "Unable to update alert status. The database may be temporarily unavailable. Please retry in a few moments." instead of "Error: database write failed").
Rationale: Generic or blaming error messages do not support efficient problem resolution and can erode operator trust during high-stress situations.

UH-UP2. The system shall provide confirmation feedback for all state-changing operations (e.g., alert acknowledgment, rule creation, subscription update) within 2 seconds of the action being submitted.
Rationale: Users require feedback to confirm their actions have been registered. A 2-second window aligns with Nielsen's response time thresholds for system acknowledgment of user input [8].

UH-UP3. All system messages shall be written in plain language without unnecessary technical jargon, accessible to operators who are not software engineers.
Rationale: City operators are domain experts in environmental management, not software. Technical language (e.g., "query returned with no matches in the database") creates unnecessary confusion.

5.2.5 Accessibility Requirements

UH-A1. All interactive elements on the operator dashboard shall be navigable using keyboard input alone, without requiring a mouse.
Rationale: Keyboard accessibility is a core requirement of WCAG 2.1 Level AA [7] and supports operators who rely on assistive technologies. In control room settings, keyboard-only workflows can also be faster for experienced operators.

UH-A2. All non-text elements (maps, charts, gauges, icons) shall include descriptive alternative text or ARIA labels for screen reader compatibility.
Rationale: Ensures the dashboard is usable by visually impaired operators relying on screen readers, in compliance with WCAG 2.1 [7].

5.3 Performance Requirements

5.3.1 Speed and Latency Requirements

PR-SL1. The alerting engine shall evaluate incoming sensor telemetry against all active alert rules and, upon detecting a threshold violation, generate and log the corresponding alert within 5 seconds of the telemetry message being received by the ingestion module.
Rationale: Environmental conditions change the order of minutes, so a 5-second end-to-end latency from ingestion to alert generation is sufficient for timely operator response while accounting for the validation, storage, and rule evaluation pipeline.

PR-SL2. The operator dashboard shall render the initial view (including map, current metrics, and active alert list) within 3 seconds of successful authentication, measured on a standard broadband connection (10 Mbps).
Rationale: The 30-second acknowledgment target (UH-EOU1) includes login and navigation time. If the dashboard takes longer than 3 seconds to render, the remaining time budget for alert acknowledgment becomes impractically constrained.

PR-SL3. The public REST API shall respond to valid requests for aggregated environmental data within 500 milliseconds (p95) under normal operating conditions.

Rationale: A 500 millisecond p95 response time is consistent with industry expectations for public-facing REST APIs. Digital signage clients and third-party integrations require timely updates to display current conditions.

PR-SL4. The telemetry ingestion module shall process and persist in a validated sensor message within 2 seconds of receipt via the MQTT broker.
Rationale: The alerting engine (PR-SL1) and dashboard depend on telemetry being available in the database promptly. A 2-second ingestion-to-storage target ensures the 5-second alerting budget remains achievable.

5.3.2 Safety-Critical Requirements

PR-SC1. The system shall not suppress, discard, or silently fail to process any telemetry message that has been validated and accepted by the ingestion module. All validated telemetry shall be durably persisted before acknowledgment to the MQTT broker.
Rationale: Environmental monitoring data informs public health decisions. Silently losing validated data could result in missed threshold violations, with potential consequences for public health. Durable persistence before acknowledgment ensures at-least-once delivery semantics.

PR-SC2. The alerting engine shall not fail to evaluate any validated telemetry message against active alert rules. In the event of an evaluation failure, the system shall log the failure and flag the affected data for manual review.
Rationale: A missed alert evaluation is functionally equivalent to a missed alert. For hazardous conditions (e.g., $PM_{2.5}$ exceeding 100 µg/m^3), failing to trigger an alert could delay protective actions [1].

PR-SC3. The system shall validate all incoming MQTT messages and API inputs to prevent injection attacks or malformed payloads from compromising system integrity.
Rationale: IoT systems are exposed to potentially untrusted data sources. Input validation at every entry point prevents malicious payloads from corrupting the database or triggering unintended system behaviour.

5.3.3 Precision or Accuracy Requirements

PR-PA1. The telemetry validation module shall reject any incoming sensor message that: (a) does not conform to the defined JSON schema, (b) contains values outside the plausible range for the corresponding sensor type, or (c) contains a timestamp deviating by more than 5 minutes from the server's current time.
Rationale: The 5-minute timestamp tolerance accounts for IoT clock drift while preventing replay attacks. Plausible value ranges (e.g., temperature between 50°C and 60°C, $PM_{2.5}$ between 0 and 1000 µg/m^3) prevent sensor malfunctions from corrupting aggregated metrics.

PR-PA2. Real-time aggregation calculations (5-minute averages, hourly maximums) shall be computed with floating-point precision of at least 2 decimal places and shall include only validated telemetry data points.
Rationale: Two decimal places are sufficient for AQI reporting standards [5]. Including only validated data prevents spurious outliers from distorting zone-level metrics.

PR-PA3. The public API shall report AQI values consistent with the EPA AQI calculation methodology [5], using the most recent complete aggregation window for the requested zone.
Rationale: AQI is a standardized metric with an established calculation methodology. Deviating from the EPA standard would produce values incomparable with other reporting systems, confusing public users and third-party developers.

5.3.4 Reliability and Availability Requirements

PR-RA1.
The system shall target an availability of 99.5% measured monthly, equivalent to no more than approximately 3.6 hours of unplanned downtime per month.
Rationale: Environmental monitoring requires continuous operation. A 99.5% target is achievable on cloud-free tiers while reflecting the expectation that the system should be available for most of each month. This is appropriate for a non-life-safety monitoring system.

PR-RA2.
Scheduled maintenance windows shall not exceed 30 minutes and shall be announced to authenticated operators at least 24 hours in advance via the dashboard.
Rationale: Predictable downtime allows operators to arrange alternative monitoring procedures. A 30-minute window limits the period during which environmental data is not being processed.

5.3.5 Robustness or Fault-Tolerance Requirements

PR-RFT1. If the connection to the time-series database is temporarily unavailable, the telemetry ingestion module shall buffer incoming validated messages in memory (up to a configurable limit) and persist upon reconnection without data loss.
Rationale: Database connectivity interruptions are a common failure mode. Buffering prevents data loss during brief outages and supports PR-SC1 (no silent data loss). The configurable limit prevents unbounded memory consumption during extended outages.

PR-RFT2. Failure of any single module (telemetry ingestion, alerting engine, data access, or security) shall not cause cascading failure of the remaining modules. Each module shall degrade independently and report its health status to the operator's dashboard.
Rationale: The project specification identifies four distinct modules. A failure in the alerting engine should not prevent the ingestion module from continuing to accept and store telemetry. Independent degradation preserves partial functionality during component failures.

PR-RFT3. The system shall handle malformed MQTT messages by rejecting them with an appropriate error response and logging the event, without impacting the processing of subsequent valid messages.
Rationale: Individual sensors may malfunction or transmit corrupted data. The system must be resilient to invalid input and continue normal operation for all other data sources.

5.3.6 Capacity Requirements

PR-C1. The telemetry ingestion module shall support concurrent data streams from at least 500 simulated sensors, each transmitting at an interval of once per minute (approximately 8.3 messages per second sustained throughput).
Rationale: A city district might deploy 50–100 sensors across zones for air quality, noise, temperature, and humidity. 500 sensors is a reasonable simulation target and produces throughput within the capacity of a single MQTT broker instance on free-tier cloud infrastructure.

PR-C2. The operator dashboard shall support at least 20 concurrent authenticated sessions without degradation of the rendering or refresh performance specified in **PR-SL2**.
Rationale: A mid-sized operations center might have 5–15 operators per shift, plus supervisors and administrators. 20 concurrent sessions provide sufficient headroom for peak usage.

PR-C3. The public REST API shall support at least 100 requests per second while meeting the response time target specified in **PR-SL3**.
Rationale: The rate-limiting controls required by the project specification [3] will enforce this capacity ceiling and protect against denial-of-service scenarios. 100 req/s accommodates a moderate number of concurrent third-party consumers and digital signage clients.

5.3.7 Scalability or Extensibility Requirements

PR-SE1. The system architecture shall support horizontal scaling of the telemetry ingestion module and alerting engine (i.e., deploying additional instances) to handle increased sensor volume without changes to the module's internal code.
Rationale: Horizontal scalability is the standard approach for IoT ingestion pipelines and avoids architectural rework if capacity requirements increase.

PR-SE2. The system shall support the addition of new environmental sensor types (e.g., UV index, wind speed) by extending the telemetry schema and validation rules, without modifying the ingestion pipeline core processing logic.
Rationale: The project specification defines four environmental indicators. A scalable system should accommodate new sensor types as monitoring needs expand. Schema-driven validation ensures that adding a sensor type is a configuration change, not a code change.

5.3.8 Longevity Requirements

PR-L1. The system shall be built using open-source technologies and standard protocols (MQTT, REST, TLS) that are actively maintained and widely adopted, to minimize the risk of dependency obsolescence over a 5-year operational horizon.
Rationale: We imagined stakeholders (a municipal government) would expect the system to remain operable for several years. A 5-year horizon is typical for municipal technology procurement cycles.

PR-L2. The time-series database schema shall support data retention of at least 12 months of raw telemetry and 5 years of aggregated data without requiring schema migration.
Rationale: Historical data is required for trend analysis and regulatory reporting. 12 months of raw data support operational analysis, while 5 years of aggregated data align with typical municipal record-keeping requirements.

5.4 Operational and Environmental Requirements

5.4.1 Expected Physical Environment

OE-EPE1. The system will operate continuously in a cloud-based environment to support geographically distributed IoT sensor devices operating in outdoor urban conditions with intermittent connectivity and variable network latency.
Rationale: System must account for and be able to tolerate network instability since sensors will be deployed in locations with constrained networks, a cloud-based environment with high volume telemetry ingestion via MQTT will help with efficient communication between devices.

5.4.2 Requirements for Interfacing with Adjacent Systems

OE-IA1. The system should be able to interface with external IoT devices using MQTT protocol over encrypted channels and should be able to send and receive information through RESTful API endpoints using HTTPS for alert notifications and public data access, while supporting rate limiting controls to prevent misuse.
Rationale: Adjacent systems include MQTT for telemetry ingestion allowing for efficient communication between devices and a read only REST API is needed for requesting and receiving of server resource from client to server endpoints for public and third-party developers.

5.4.3 Productization Requirements

OE-P1. Upon launch of the product, the system should boast several distinguishable features such as role-based access control (RBAC) for administrative usage, configurable alert rules, documented APIs

for public and third-party usage, a functional client, encrypted data using industry standard protocols and a secure dashboard for city operators.
Rationale: These are all the expected features provided from the system capabilities documentation that stakeholders would expect the system to have at product launch.

5.4.4 Release Requirements

OE-R1. The system must be deployable on a standards-compliant cloud, support secure MQTTs or TLS and HTTPS communication, alongside being compatible with modern web browsers such as Chrome, Firefox or Edge.
Rationale: Browser compatibility guarantees that operators can acknowledge critical environmental alerts within the targeted 30 second time frame, alongside being accessible to all users.

5.5 Maintainability and Support Requirements

5.5.1 Maintenance Requirements

MS-M1. The system shall provide automated build and test execution on every merge to the main branch and shall block deployment if automated checks fail.
Rationale: Continuous integration reduces the probability of introducing defects into core pipelines (telemetry ingestion and alert generation) where reliability is critical.

MS-M2. The system shall maintain versioned database schema migrations and shall support forward-only upgrades without manual database edits.
Rationale: Telemetry storage and aggregation are core capabilities; repeatable migrations reduce operational risk and prevent inconsistent environments.

5.5.2 Supportability Requirements

MS-S1. The system shall provide auditability for support investigations by retaining: (a) authentication logs, (b) administrative action logs, and (c) alert lifecycle logs (triggered/acknowledged/resolved) for at least 90 days.
Rationale: Support teams need historical traces to reproduce issues and explain incidents; audit logs are already a core security expectation in the project.

MS-S2. The system shall include runbooks for the top operational incidents: (a) MQTT ingestion failure, (b) database connection failure, (c) alerting engine evaluation failure, and (d) public API latency degradation. Each runbook shall include detection symptoms, likely causes, and step-by-step mitigation actions.
Rationale: Operators need predictable recovery steps when core pipelines fail; runbooks reduce mean time to restore.

5.5.3 Adaptability Requirements

MS-A1. The public REST API shall be versioned (e.g., /v1/...) and shall maintain backward compatibility within a major version; breaking changes shall require a new major version and a published deprecation window of at least 60 days.
Rationale: Third-party developers depend on stable endpoints; versioning supports evolution without breaking integrations.

MS-A2. The telemetry validation layer shall support adding new sensor types by extending schema/validation rules without modifying the ingestion pipeline's core message-processing flow.
Rationale: The project explicitly anticipates growth in sensor indicators and emphasizes schema validation and scalable ingestion.

5.6 Security Requirements

5.6.1 Access Requirements

SR-AC1. The system will require all operators and administrators to authenticate and log in before using any internal features.
Rationale: Authentication ensures that only authorized users can access operational and administrative features of the system.

SR-AC2. The system will enforce role-based access control (RBAC), so each user can only see features and data that match their specific role.
Rationale: Different roles have different responsibilities and privileges. RBAC prevents unauthorized access to sensitive functions such as configuration changes or audit log access.

SR-AC3. The system shall deny access to protected resources by default unless explicit permission is granted to the authenticated role.
Rationale: A default-deny access policy reduces the risk of accidental privilege escalation and limits the impact of misconfiguration.

5.6.2 Integrity Requirements

SR-INT1. All incoming telemetry data and API requests shall be protected against unauthorized modification using transport-layer security (TLS).
Rationale: Environmental monitoring decisions rely on the accuracy of received data. Protecting data integrity ensures that sensor readings and API responses are not altered in transit or by malicious actors, preserving trust in system outputs.

5.6.3 Privacy Requirements

SR-P1. The system shall minimize the collection and storage of personal data and shall not collect personally identifiable information from public users.
Rationale: Environmental monitoring does not require personal data. Minimizing data collection reduces privacy risks and regulatory exposure.

SR-P2. Operator and administrator account information shall be accessible only to authorized users and shall not be exposed through public APIs.
Rationale: Restricting access to account data protects user privacy and reduces the risk of targeted attacks.

SR-P3. Audit logs and operational data shall not be publicly accessible and shall be viewable only by authorized administrative roles.
Rationale: Audit and operational data may contain sensitive system information that should not be exposed to unauthorized users.

5.6.4 Audit Requirements

SR-AU1. The system shall log all authentication attempts, including successful and failed login events.
Rationale: Authentication logs support the detection of unauthorized access attempts and potential security incidents.

SR-AU2. The system shall record administrative actions, including configuration changes and user role modifications.
Rationale: Recording administrative actions supports accountability and post-incident investigation.

SR-AU3. Audit logs shall include timestamps and identifiers sufficient to trace actions to individual authenticated users.

Rationale: Detailed audit records enable effective forensic analysis and operational oversight.

5.6.5 Immunity Requirements

SR-IM1. The system shall reject malformed or unexpected telemetry messages and API requests without impacting the processing of valid inputs.
Rationale: Rejecting malformed input protects the system from crashes and ensures continued operation under adverse conditions.

SR-IM2. The system shall implement rate-limiting controls on public and authenticated APIs to mitigate denial-of-service attacks.
Rationale: Rate limiting prevents excessive request volumes from degrading system availability.

5.7 Cultural and Political Requirements

5.7.1 Cultural Requirements

CP-C1. The system shall not display environmental alerts, messages, or symbols that may be considered offensive to any religion, ethnicity, disability group, sexual orientation, or demographic group within the city.
Rationale: Residents must feel safe and respected when interacting with the system to encourage adoption and trust.

CP-C2. The system shall provide environmental alerts and safety information in languages commonly used within the city population.
Rationale: Providing alerts in commonly used languages ensures that environmental safety information can be understood by a broader population.

CP-C3. The system shall allow users to report environmental hazards, system misuse, or inappropriate content through the platform.
Rationale: Allowing users to report issues encourages responsible system use and improves overall system reliability and safety.

CP-C4. The system shall provide environmental alert information in formats accessible to individuals with disabilities.
Rationale: Individuals with disabilities must be able to receive and understand environmental alerts to ensure equal access to public safety information.

5.7.2 Political Requirements

CP-P1. The system shall operate in compliance with applicable government regulations related to environmental monitoring and public safety communication.
Rationale: Compliance with regulations is required for lawful operation and public sector deployment.

CP-P2. The system shall protect user and environmental data from unauthorized access or misuse.
Rationale: Protecting data maintains public trust and supports legal and ethical operations.

CP-P3. The system shall support authorized government agencies in issuing environmental emergency alerts to the public.
Rationale: Government agencies must be able to distribute urgent safety information during environmental emergencies.

5.8 Legal Requirements

5.8.1 Compliance Requirements

LR-COMP1. All personally identifiable information (PII) collected by the system must be protected from unauthorized access, disclosure, or misuse.
Rationale: This aligns with PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 7 - Safeguards [18].

LR-COMP2. The purposes for collecting personally identifiable information (PII) must be identified before or at the time the information is collected.
Rationale: This aligns with PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 2 - Identifying Purposes [19].

LR-COMP3. The knowledge and consent of the user must be obtained before collecting, using, or disclosing personally identifiable information (PII), except where required by law.
Rationale: This aligns with PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 3 - Consent [20].

LR-COMP4. The system shall limit the collection of personally identifiable information (PII) to only what is necessary to fulfill identified system purposes.
Rationale: This aligns with PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 4 - Limiting Collection [21].

LR-COMP5. Personally identifiable information (PII) collected by the system shall only be used, retained, or disclosed for identified purposes unless otherwise required by law or authorized by the user.
Rationale: This aligns with PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 5 - Limiting Use, Disclosure, and Retention [22].

5.8.2 Standards Requirements

LR-STD1. The system shall follow recognized public emergency alert communication standards when delivering environmental alerts.
Rationale: This follows the Common Alerting Protocol (CAP), which defines standards for emergency alert message structure and delivery [23].

LR-STD2. The system shall comply with recognized accessibility standards for digital public services.
Rationale: This follows the Web Content Accessibility Guidelines (WCAG), which provide standards for accessible digital interfaces [24].

LR-STD3. The system shall comply with recognized information security standards to protect system data and communications.
Rationale: This follows ISO/IEC 27001, which defines standards for information security management systems [25].

LR-STD4. The system shall follow recognized requirements of engineering and data management standards during system development and operation.
Rationale: This follows ISO/IEC/IEEE 29148, which defines standards for requirements engineering and system lifecycle processes [26].

6 Innovative Feature

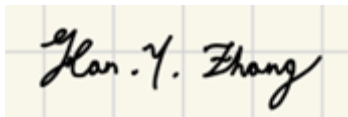
The chosen innovative feature to implement was personalized alert subscriptions for city operators. This feature allows operators to subscribe to specific environmental metrics, zones, or alert types that are most relevant to their responsibilities. By receiving only, the alerts that matter to them, operators can reduce alert fatigue, respond more efficiently, and maintain better situational awareness. This feature enhances the usability of the dashboard and improves operational efficiency without adding significant system complexity. Some other innovative features are listed below:

- Zone-based environmental profiles (users save preferred zones for quick monitoring) [9]
- Accessibility Dashboard Mode (options for text size and contrast adjustments) [10]
- Environmental Impact Score (simple score summarizing overall zone quality) [11]
- Shareable Environmental Snapshots (export current conditions for other teams) [12]
- Premium Analytics Subscription (advanced insights and extended historical data for public users) [13]
- Operator Performance Ranking (scores city operators based on alert response speed) [14]
- Smart Alert Bundling (combines related alerts into one notification) [15]
- Environmental Event Timeline (shows how conditions changed before an alert) [16]
- Third-Party Integration Marketplace (list of approved third-party tools, i.e., weather services, traffic data providers, that can integrate with the system's API) [17]

A Division of Labour

Zhang, Han:

- Section 3 Use Case Diagram
- Section 4.2 BE2 Request and Interpret Environmental Data
- Section 5.4 Operational and Environmental Requirements
- Helped with the conceptualization of business events
- Helped with the conceptualization of modules and functions



Pham, Aaron:

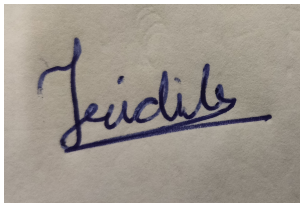
- Section 1 (rework 1.3)
- 5.1 (independently)
- 5.2 (independently)
- 5.3 (independently)
- Section 5 and 1 Format

Anura, Mithu:

- Section 2.1: Production Perspective
- Section 2.1: Block diagram
- Section 2.3: User Characteristics
- Section 2.4: Constraints
- Section 2.5: Assumptions and Dependencies
- Section 2.6: Apportioning of Requirements
- Section 5.6: Security Requirements

Khorramijam, Arsia:

- Brainstormed Business Events; created ideas for B1-B4
- Section 4: BE3
- Section 4: BE4
- Section 5.5; Maintenance, Support, Adaptability

Banik, Tridib:

- 1.3 Definitions, Acronyms, and Abbreviations
- 2.2 Product Functions; first paragraph and state diagram are done individually, but the table (list of modules and functions) is done as a group
- 4 Highlights of Functional Requirements: BE1 Authentication and BE6 Manage Personalized Alert Subscriptions are done individually, but the list of all business events and the list of viewpoints are done as a group
- 6 Innovative Features; brainstormed all innovative features with brief descriptions (in round brackets) for each of them
- 5.7 Cultural and Political Requirements
- 5.8 Legal Requirements
- Transferred all content from Word document to Overleaf LaTeX, properly formatted the entire document, including spacing, indexing, justified paragraphs, bolded text, positioning of bullet points or numbers, etc.